The Paleomagnetism of Kaminak Dikes—No Evidence of Significant Hudsonian Plate Motion

K. W. CHRISTIE, A. DAVIDSON, AND W. F. FAHRIG
Geological Survey of Canada, 601 Booth Street, Ottawa, Canada K1A 0E8

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A paleomagnetic study was done on 28 diabase and 6 lamprophyre dikes from the vicinity of Kaminak Lake in the District of Keewatin. These dikes are Proterozoic and cut across an area of Archean rocks. Their metamorphic grade within this region varies from nil to amphibolite facies. Two stable magnetization directions were obtained from the diabase on AF demagnetization; one from 3 unmetamorphosed diabase dikes of D = 205°, I = 3°, α_{95} =24°, with a corresponding pole position at 23.8° S, 122.3° W, α_{95} = 16.5°; the other from the metamorphosed diabase of D = 176°, I = 65°, α_{95} = 4°, with pole position at 20.4° N, 92.1° W, α_{95} = 6.1°. The first is believed to be thermoremanent and approximately 2300 Ma old, while the second is a metamorphic direction of approximately 1800 Ma. The six lamprophyres give a stable remanence of D = 174°, I = 78°, α_{95} = 10°, with a pole position at 40.9° N, 92.8° W, α_{95} = 17°, essentially the same as the metamorphosed diabase.

A number of samples were also thermally demagnetized. Most of the results obtained were the same as those obtained from the AF treatment. However, one diabase site gave a direction similar to that obtained from the 3 unaltered dikes, and one of the lamprophyres showed a reversal.

The pole position of the altered Kaminak diabase relative to that for the unconformably overlying Dubawnt Group suggests that the polar wandering curve for this part of the Aphebian may be more complex than previously thought. It is quite probable that the earlier Aphebian pole positions (Matachewan, Nipissing), as usually plotted, are actually antipodal points in terms of the polar wandering curve.

Une étude paléomagnétique fut poursuivie sur des dykes de diabase (28) et de lamprophyre (6) dans la région du Lac Kaminak dans le District de Keewatin. Ces dykes protérozoiques traversent des roches archéennes. Leur degré de métamorphisme dans cette région varie de nul à faciès d'amphibolites. Deux directions stables de magnétisation ont été obtenues des diabases par la méthode de démagnétisation S.F.; une des directions provient de 3 dykes non métamorphisés de D = 205°, I = 3°, α_{95} = 24° avec la position du pôle correspondant à 23.8° S, 122.3° W, α^{95} = 16.5° l'autre direction provient des diabases métamorphisés de D = 176°, I = 65°, α^{95} = 4° avec la position du pôle à 20.4° N, 92.1° W, α^{95} = 6.1°. On croit que la première magnétisation est de type thermorémante et date de 2.300 Ma, la seconde est une direction magnétique datant d'environ 1.800 Ma. Les six lamprophyres donnent une rémanence stable de D = 174°, I = 78°, α^{95} = 10° avec la position du pôle à 40.9° N, 92.8° W, α^{95} = 17° essentiellement la même que celle des diabases métamorphisés.

Un certain nombre d'échantillons avaient subi une démagnétisation thermique. La plupart des résultats obtenus étaient les mêmes que ceux de la méthode AF. Cependant, l'un des diabases donnait une direction semblable à celle obtenue pour les trois diabases non altérés et un des lamprophyres montrait un magnétisme inverse.

Les données paléomagnétiques favorisent l'hypothèse de l'intrusion des lamprophyres pendant l'étape tardive de l'orogénie Hudsonienne (thermique) dans cette région.

La position du pôle du diabase altéré de Kaminak, par rapport au pôle du Groupe de Dubawnt sus-jacent et discordant, suggère que la courbe de migration du pôle pour cette partie de l'Aphébien pourrait être plus complexe qu'on ne le croyait jusqu'à présent. Il est probable que les positions du pôle à l'Aphébien (Matachewan, Nipissing) telles qu'elles étaient tracées auparavant sont en fait des points antipodaux par rapport à la courbe de migration des pôles.

[Traduit par le journal]

Introduction

A number of discrete areas of Archean rocks occur within the confines of the Canadian Shield. They are separated from one another by rocks that have been involved in Hudsonian or later orogenies. The original spatial relationships of the Archean areas and the question of whether movements have taken place among them since the Kenoran Orogeny are of considerable interest. Paleomagnetic studies of

Early Aphebian rocks that intrude or lie undisturbed on these Archean blocks will contribute to the understanding of any movements that have taken place between these blocks since the Early Proterozoic.

The Kaminak dikes are early Proterozoic and cut one such Archean area (the Kaminak Orogen) that lies isolated within the Churchill Orogen (Fig. 1). Their paleomagnetism has been examined to see if it will throw some light on the question of whether this Archean area has moved relative to, for example, the Superior Orogen, during or since the Hudsonian Orogeny.

Regional Setting

The Churchill Structural Province is defined as that part of the Canadian Shield in which the latest and generally predominating structure and metamorphism were produced by the Hudsonian Orogeny, considered to have ended about 1735 Ma (million years) ago (Stockwell 1970, p. 49). Within this Province, the Kaminak Subprovince (Davidson 1972) is defined as an area in which older rocks, structures, and metamorphism are well preserved, an 'island' that has escaped the obliterative effects of Hudsonian Orogeny. The Kaminak Subprovince extends for several hundred kilometres west–southwest of Hudson Bay and is as much as 110 km wide. It is bordered to the north and south by high-grade metamorphic terrains.

Stratigraphic Succession and Age

The oldest rocks in the Kaminak Subprovince are mafic and felsic volcanic rocks, graywacke-turbidites, and banded iron-formation of the Kaminak Supergroup. These rocks have been steeply folded about major east–west axes and mildly regionally metamorphosed. After a long period of erosion, a sedimentary succession locally including minor volcanic rocks, was deposited. These rocks comprise the Hurwitz Group (Wright 1955; Bell 1970). The Hurwitz Group and the basement rocks on which they lie are severely deformed north and south of the Kaminak Subprovince, and grade into gneisses. Some 160 km to the north, sedimentary and volcanic rocks of the Dubawnt Group overlie these gneisses.

The Kaminak Supergroup and the pre-Hurwitz plutonic complexes are Archean, giving K–Ar hornblende, Rb–Sr isochron, and U–Pb sphene and zircon ages in the range 2.5 to 2.7 Ga (billion years) (Wanless *et al.* 1973; Wanless and Eade 1975); K–Ar biotite ages are variously up-dated. These rocks are essentially identical to those of the greenstone belts of the Superior Structural Province (Goodwin 1972). The Hurwitz Group is Aphebian, but the time spanned by the group is not completely established. Recent ages determined on volcanic flows within, and tuffaceous metasediments correlated with, the Hurwitz Group are of the order of 1800 (1910)[1] Ma, or late Aphebian (Wanless and Eade 1975). Volcanic rocks in the post-Hudsonian Dubawnt Group are 1725 (1828) $\pm$ 4 Ma old (Wanless *et al.* 1972, p. 25).

Dikes

Three main ages of dike intrusion are known within the Kaminak Subprovince, and all are represented within the area of this study (Fig. 1). From oldest to youngest, these are: (1) northerly trending porphyritic Kaminak diabase dikes that cut all the Archean rocks but are truncated by the unconformity at the base of the Hurwitz Group; (2) east- to southeast-trending lamprophyre dikes that cut folded Hurwitz Group rocks and the metamorphosed Kaminak dikes; and (3) northwest-trending non-porphyritic Mackenzie diabase dikes that cut the Dubawnt Group and all older rocks. In addition, a few easterly-trending non-porphyritic diabase dikes, some altered, are known to cut the Kaminak dikes in the vicinity of Kaminak Lake and segments of a variety of older dikes, presumed related to Archean volcanism, are present within the Kaminak Supergroup.

Of particular interest in this study are the northerly-trending Kaminak diabase dikes and the lamprophyres.

Kaminak Diabase Dikes

The Kaminak dikes are best exposed in the vicinity of Kaminak Lake (Fig. 1) and range from less than 1 m to rarely more than 30 m wide. Single dikes can rarely be traced for more than a few kilometres, mainly because they are segmented and laterally displaced by faults across which they cannot be identified. The dikes

[1]Throughout this paper Rb–Sr isochron ages are reported in pairs, based on the alternate decay constants 1.47 and 1.39 $\times$ 10^{11} y^{-1}. The age derived from the latter constant is parenthesized.

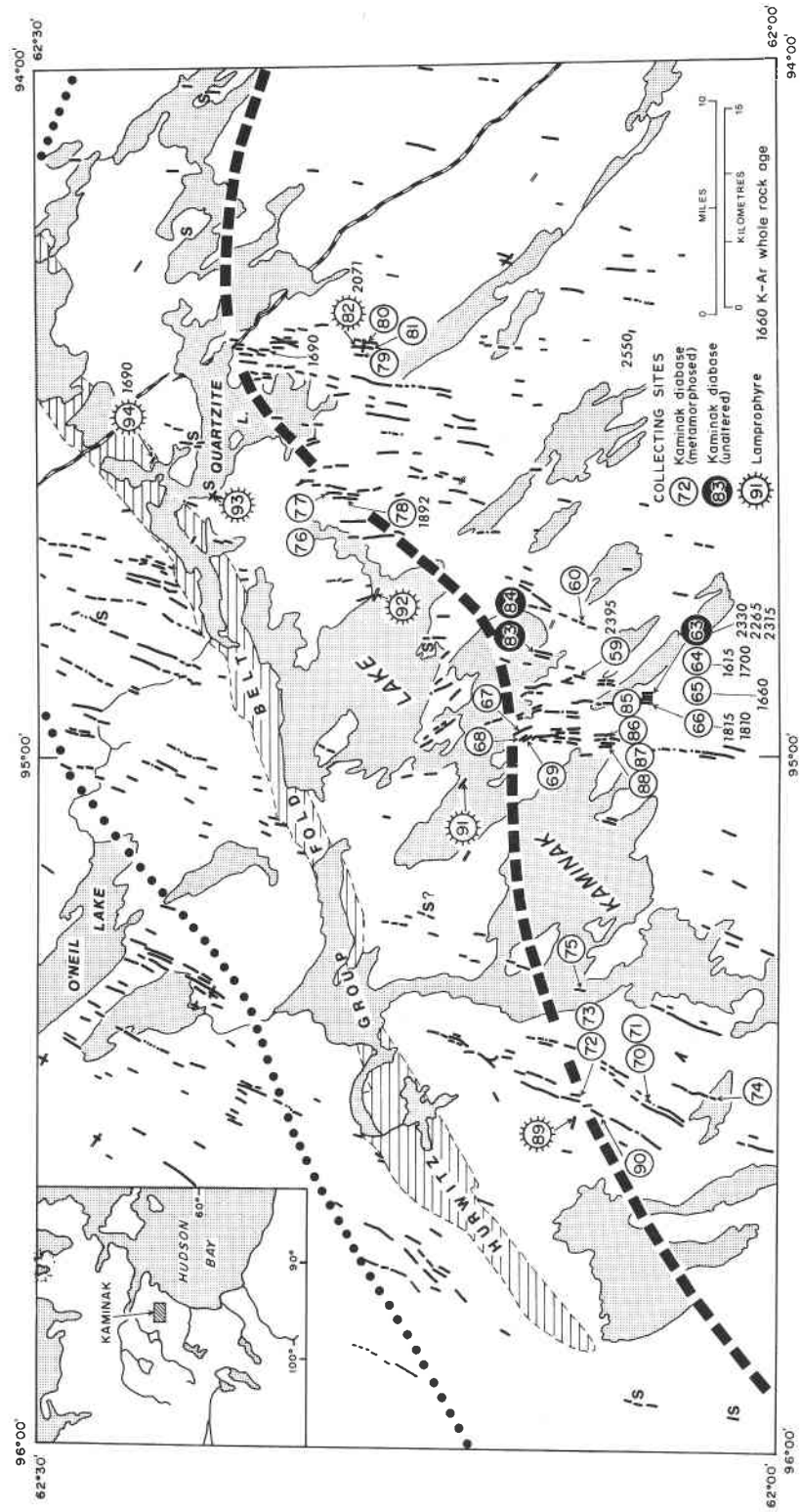


FIG. 1. Distribution of dikes and collecting sites, Kaminak Lake vicinity. Solid line = Kaminak diabase, double line = lamprophyre, patterned dike = Mackenzie diabase. Some Kaminak diabases contain stilpnomelane (S) north of heavy dashed line. Kaminak diabase is metamorphosed to amphibolite north of dotted line.

have no expression on aeromagnetic maps. The characteristic plagioclase phenocrysts range in size from a few millimetres to 5 cm or more. Generally they constitute a very few percent of the diabase, but locally they are crowded into zones parallel to the dike walls.

Within the central part of the Kaminak Subprovince, the diabase dikes range from entirely fresh to partly or wholly altered. The fresh diabase dikes have dark, charcoal gray, chilled margins with microphenocrysts of clear plagioclase and augite, and dark gray, medium grained cores with ophitic texture and glassy plagioclase phenocrysts. In thin section, clear plagioclase laths are seen to be partly enclosed by both augite and pigeonite. Early crystallized olivine is usually present in small amounts. Thin rims of brown hornblende border pyroxene grains locally. Quartz is absent. Opaque oxide grains are chiefly Ti–magnetite–ilmenite intergrowth; skeletal ilmenite grains are uncommon.

In contrast, the more completely altered diabase dikes have greenish gray chilled margins and darker greenish gray, medium grained cores with clouded, dull white or yellow plagioclase phenocrysts. Thin sections reveal a variety of alteration products, although the original ophitic texture is usually preserved. Groundmass plagioclase is commonly sericitized and associated with secondary epidote. The original pyroxenes are changed to uralitic amphibole with colorless cores and brownish outer zones; discontinuous rims of blue-green hornblende are present locally. Clots of very fine grained chlorite and antigorite dusted with tiny opaque grains are pseudomorphous after olivine. Patches of fine sphene (leucoxene) have replaced the opaque oxide grains and commonly preserve the lamellar form of Ti–magnetite–ilmenite intergrowths. Secondary brownish biotite is commonly associated with altered opaque oxide grains. Fine epidote, chlorite, greenish biotite, and minor pyrite, calcite, and quartz intergrown with albite are present interstitially. Neither prehnite nor pumpellyite have been identified.

In the central part of the Kaminak Subprovince, the distribution of fresh and altered dikes is apparently haphazard. Both fresh and altered dikes may occur in the same outcrop area, and fresh dikes may become altered along strike. To the north and south all the dikes become altered, and can be traced through zones of increasingly higher-grade metamorphic assemblages and structural degradation to where they are reduced to large boudins of amphibolite within migmatitic gneisses. The heavy dashed line in Fig. 1 delimits the first obvious metamorphic effect; primary pyroxene has not been found in thin sections of diabase north of this line, nor has the metamorphic mineral stilpnomelane been found south of it. Sampling for the current paleomagnetic study was restricted to the area in Fig. 1 south of this line. Although many of the diabase dikes south of this line are altered, it seems likely that some, if not most, of this alteration was the result of deuteric hydration during original cooling, rather than being solely an effect of Hudsonian metamorphism. The occurrence of secondary hydrous minerals in the altered diabases that would form only at higher metamorphic grade than the first appearance of stilpnomelane is evidence in this regard. It is considered that previously altered (hydrated) diabase would be more susceptible to change during subsequent low grade metamorphism than would fresh (dry) diabase. Alternatively, much of the lower grade alteration of the diabases may have been caused by Hudsonian metamorphism, with local variations in the amount of available water allowing parts of some of the dikes to remain unaltered.

Twelve K–Ar whole-rock ages have been determined on chilled margin samples of eight dikes (Wanless *et al.* 1973, pp. 42–48). Eight of these ages are from one locality where four parallel dikes lie close together (Fig. 1, sites 63 to 66). Three of the eight dated dikes are fresh, and the five ages (three from one dike) are all greater than 2250 $\pm$ 200 Ma, the average of these five ages being 2370 $\pm$ 200 Ma. The other five dated dikes are variously altered and yield ages ranging from 1615 to 1892 Ma (error limits between $\pm$ 66 and $\pm$ 185 Ma). The greatest of these ages (Fig. 1, site 78) was obtained from a partly altered dike in which cores of primary pyroxene are preserved in the medium-grained central part of the dike. The early Aphebian age of the fresh dikes establishes a maximum age for the Hurwitz Group. It is similar to the 2485 Ma age reported for the Matachewan diabase dikes in the Superior Province, which are also porphyritic, north trending, and cut Archean plutonic and east-trending supracrustal rocks (Fahrig *et al.* 1965). The range of ages obtained from the altered dikes is attributed to variable radiogenic argon loss during or since the

Hudsonian Orogeny. This variability may be due to greater susceptibility for argon loss from previously altered dikes than from fresh ones, or it may be due to variation in the amount of low grade recrystallization during the Hudsonian.

Lamprophyre Dikes

Gray lamprophyres with biotite phenocrysts occur as narrow dikes, generally less than 2 m wide, throughout the region from Hudson Bay to Lake Athabaska. In the Kaminak Subprovince, the lamprophyre dikes trend between north 70° east and south 60° east. They are commonly sinuous in detail, or form very narrow swarms of discontinuous dikelets. In several places they are full of rounded country rock xenoliths. In thin sections, the lamprophyre is seen to be composed primarily of K-feldspar and biotite with lesser amounts of quartz, albite, and pale amphibole, and accessory sphene, apatite, and calcite. The larger biotite phenocrysts have colorless (phlogopitic?) cores and orange-brown frayed rims, and are commonly bent.

Lamprophyre dikes cut all rock units within the area, with the exception of the northwest-trending Mackenzie diabase dikes. Unmetamorphosed lamprophyre dikes cut cleanly across metamorphosed Kaminak dikes, now amphibolites with granular plagioclase augen in place of primary phenocrysts, in the vicinity of O'Neil Lake (Fig. 1). This relationship suggests that the intrusion of lamprophyre was at least in part post-Hudsonian. Whether or not they are related to Dubawnt igneous activity (Davidson 1970) remains unproven.

Sampling and Measurement

One hundred and ninety-four cores were drilled in the field and oriented (with solar compass) at 28 sites on Kaminak diabase dikes (Fig. 1). The 28 sites probably represent 28 separate intrusive bodies. One or more cores were collected from country rock, baked by the dike intrusions, at sites 67, 79, and 80. Thirty-eight samples were also collected from 6 lamprophyre dikes (Fig. 1). Two specimens were cut from each core and their NRM measured on a biastatic magnetometer (Larochelle and Christie 1967). Two or three test specimens from each site were then AF demagnetized in 50 oersted steps (from 50 to 350) then at 500 and 800 Oe, to determine the optimum field for the elimination of 'secondary' unstable components from each site. All of the specimens from many sites were cleaned in fields of 100, 150, and 200 Oe. From these various tests, it was concluded that the optimum AC cleaning field for all sites was 200 Oe. This was found to be true for both the diabase and lamprophyre specimens. Weakly magnetized specimens ($<2.10^{-6}$ emu/cm^3) were measured with a JR-3 spinner magnetometer (manufactured by Institute of Applied Geophysics, Brno, Czechoslovakia).

Data from cores were excluded from final statistics if: (1) the angle θ between the magnetization direction of the two specimens exceeded 25° after optimum cleaning, (2) the direction of magnetization of a core (mean direction of the two specimens from that core) differed by more than 3 angular standard deviations from the mean direction of the site, or (3) only one core remained to represent the site after applying rejection criteria (1) and (2).

Results

The site mean directions of the metamorphosed, stably magnetized, Kaminak diabase dikes are shown on Fig. 2A; the directions of unmetamorphosed sites on 2B; and the directions of the 6 lamprophyre sites on 2C. Data for sites 75, 77, and 78 are not shown on Fig. 2, as the cores from these sites did not meet the rejection criteria given above.

The mean site directions of the metamorphosed dikes are well grouped about D = 176°, I = 65°, and the three sites from unmetamorphosed dikes are less well grouped about D = 205°, I = 3° (Table 1, Fig. 2A,B). Because the dikes grade as a result of regional metamorphism from completely unaltered to intensely altered, it was expected that one direction would be obtained for the unaltered dikes, but that directions for the remainder would be streaked away from this direction. Such was not the case, as there is a clear separation between the magnetization directions of the unaltered and altered dikes. Some streaking is discernible, however. For example, one core from site 63 is stably magnetized in the metamorphic direction, whereas the remainder (6) have the gentle southwesterly (unaltered) magnetization direction.

The magnetization directions of cores from several other sites also exhibited this tendency to be streaked out between the two extremes, or

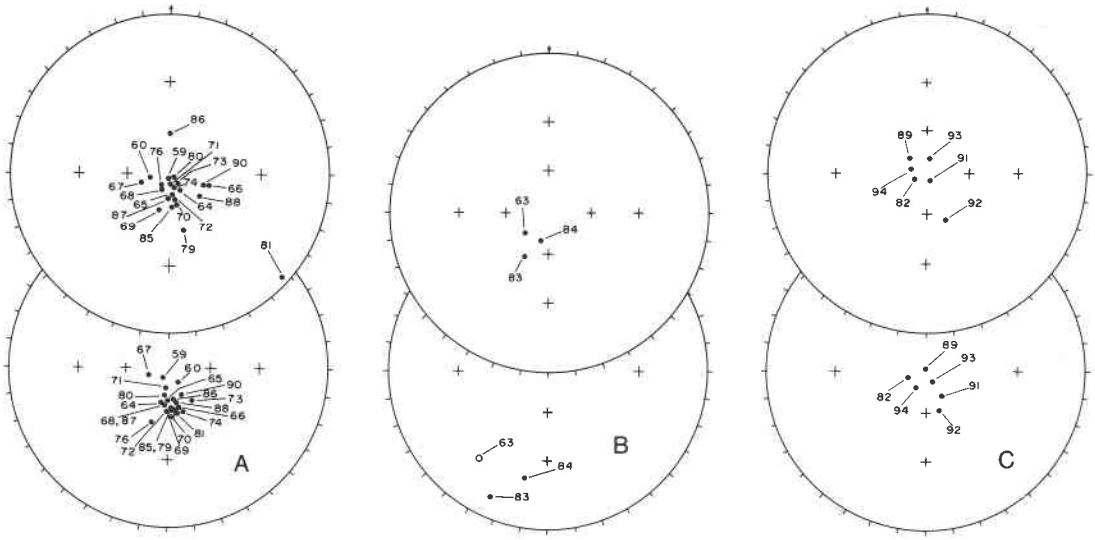


FIG. 2. Stereographic projections of site mean magnetization directions before (upper) and after (lower) optimum AF cleaning (solid dots [open circles] indicate downward [upward] north-seeking magnetization) of: A, metamorphosed diabase dikes; B, three unaltered diabase dikes; and C, six lamprophyre dikes.

exhibited other peculiar characteristics. Site 75 had only one homogeneously magnetized core after cleaning at 200 Oe, but at this, and lower fields, the directions of cores from this site were streaked towards the unmetamorphosed direction. Similarly, only one sample from site 76 had a low θ after 200 Oe treatment, and its direction agreed with that of the metamorphosed group; two of the high θ samples had an unmetamorphosed direction. The directions for site 77 were random and the cores inhomogeneously magnetized. Five samples from site 85 were tightly grouped in the metamorphic direction (at 200 Oe), but two samples (one with high θ) lay in the unmetamorphosed direction. One sample from site 65 was stably magnetized in the present earth's field direction, but the remainder were grouped in the metamorphic dike direction. Five cores from site 86 had a good metamorphic dike direction, the other two were highly divergent. Finally, 5 cores from site 70 had a good metamorphic direction; the directions of the remaining two were highly divergent and very intensely magnetized. The latter may be due to the effects of lightning strikes.

The magnetization directions of individual cores from the metamorphosed dikes, after optimum cleaning, are shown on Fig. 3. Figure 3A shows the directions of those cores (6) that were highly divergent relative to the rest of the site and the directions of 73 cores (44%) that are inhomogeneously magnetized. Figure 3B shows the directions of the 94 acceptable cores. The directions of cores in Fig. 3A (the 'unstable' metamorphosed cores) are concentrated about a direction similar to those of 3B (the stably magnetized metamorphosed cores), but those of 3A show a greater dispersion about the mean direction. A significant number of these cores appear to retain a component in the recent earth's field direction. Seven (33%) of the unmetamorphosed cores were rejected because of inhomogeneous magnetization after AF cleaning.

Baked contact samples (4) were obtained at three diabase dikes (sites). These sites are 67, 79, and 80. The diabase at these sites is metamorphosed and the cleaned directions of the homogeneously magnetized cores of sites 67 and 80 are all in the metamorphic dike direction. The two baked contact samples from these two sites (67 and 80) have similar cleaned directions. This is not a baked contact test (Everitt and Clegg 1962), as the dikes and country rock were undoubtedly remagnetized together long after the consolidation of the dikes. On the other hand, although most of the samples from

TABLE 1. Paleomagnetic data for diabase and lamprophyre dikes of the Kaminak Lake area

			NRM				Cleaned at 200 Oe				
Site	*N*	*n*	*M*	D	I	*k*	*M*	D	I	*k*	α
Kaminak Diabase Dikes (metamorphosed)											
64	7	6	71.8	152	77	8.4	28.8	189	64	64	7
65	7	6	113	172	74	59	50.4	180	66	82	6
66	6	3	1261	107	60	3.0	57.6	166	60	4.1	40
67	6	2	101	254	69	205	17.2	251	75	61	13
68	7	6	186	208	78	2.4	52.0	186	63	27	11
69	7	7	102	184	63	3.1	49.6	176	56	162	4
70	7	5	217	168	67	108	102.8	170	60	1804	1
71	7	4	81.4	173	81	286	27.8	186	74	73	8
72	7	2	98.2	169	71	130	58.8	181	59	85	11
73	7	3	30.7	152	80	56	14.5	144	61	7.3	30
74	7	5	446	158	80	697	43.9	162	58	168	4
76	7	3	111	212	79	15	8.4	197	51	6.6	32
79	6	4	259	166	49	3.2	44.0	173	59	92	7
80	8	5	302	143	86	351	60.1	188	71	50	8
81	6	2	16.1	132	2	1.0	7.9	169	57	596	4
85	7	5	171	173	66	41	111.8	174	59	732	2
86	7	5	471	3	61	1.4	111.8	170	66	159	5
87	7	4	496	180	72	11	35.9	186	63	34	12
88	7	6	28.3	126	63	4.2	12.6	163	62	18	13
90	7	3	517	109	64	2.2	34.2	153	68	27	16
59	7	3	122	193	85	11	21.5	208	82	13	23
60	9	5	285	260	76	26	41.9	150	76	12	18
Sites		22		158	75	13		176	65	55	4
Samples		94		165	76	5.5		175	65	24	2
						Mean pole position*	092.1° W, 20.4° N			$k = 24$	$\alpha = 6.1°$
Kaminak Diabase Dikes (unmetamorphosed)											
63	7	3	1198	228	68	312	131	218	−19	68	9.8
83	7	5	801	209	55	35	59.8	205	7.4	67	7.3
84	7	5	1438	196	69	35	87.2	192	20	43	9.6
Sites		3		211	65	65		205	3.0	12	24
Samples		13		209	64	33		203	6.0	15	10
						Mean pole position*	122.3° W, 23.8° S			$k = 24$	$\alpha = 16.5°$
Lamprophyre Dikes											
82	7	6	11.0	250	81	19	7.3	249	76	10	18
89	7	4	15.5	314	73	236	2.7	036	88	145	5.8
91	7	5	59.3	154	83	28	43.1	148	69	18	15
92	5	5	23.8	159	54	27	18.1	160	60	123	5.7
93	7	7	228	14	80	42	7.6	153	80	23	11
94	5	4	7.6	284	79	88	2.5	209	76	623	2.8
Sites		6		220	86	18		174	78	33	10
Samples		31		214	86	14		174	78	17	6.1
						Mean pole position*	092.8° W, 40.9°N			$k = 11$	$\alpha = 17°$

N = number of samples collected at site
n = number of samples per site after rejecting unstable samples as explained in text
M = arithmetic mean of the intensity of magnetization in emu × 10^{-6}
D = declination (degrees)
I = inclination (degrees)
k = Fisher's (1953) best estimate of precision
α = half angle (degrees) of the cone of confidence at $P = 0.95$.
*Note—Mean pole positions were calculated from site pole positions

site 79 had a cleaned direction similar to that of the metamorphosed dikes, a chilled contact sample and a baked (amphibolite) country rock sample had a cleaned direction that was steeply up in a southeasterly direction. Both samples are stably magnetized, *i.e.*, they showed good end-points and low θ's after cleaning at 200 Oe. We have no explanation for this odd direction, unless a lightning strike has affected both the chilled margin and the baked contact a few centimetres away.

The site mean directions of the 6 lamprophyre

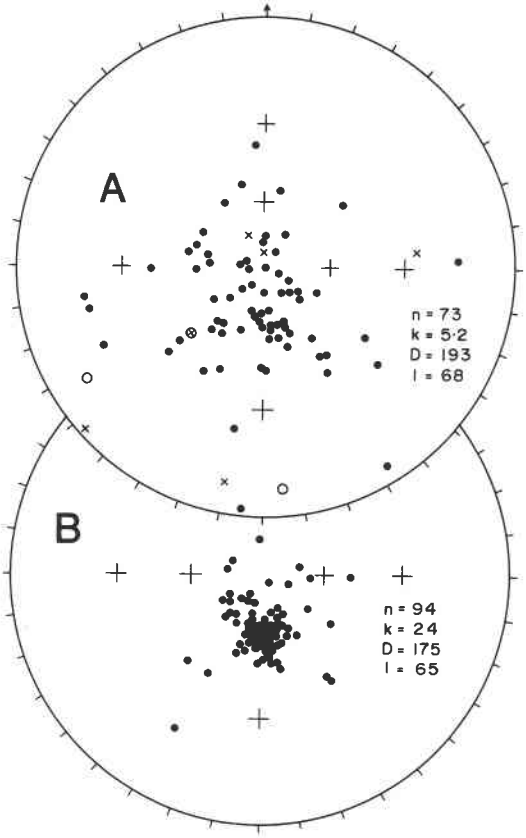


FIG. 3. Magnetization directions of 167 cores from metamorphosed dikes after optimum AF cleaning. A: Directions of cores that were either inhomogeneously magnetized (open circle—up, dots—down) or which were highly divergent relative to the remaining cores of their site (crosses—down; circled cross—up). B: Directions of acceptable cores.

dikes are given in Table 1 and shown on Figure 2C before (NRM) and after AF cleaning at 200 Oe. The statistics for these dikes are given in Table 1. Seven cores (18%) were inhomogeneously magnetized after AF cleaning. This is a notably smaller percentage than that of the metamorphosed and unmetamorphosed diabase dikes (44% and 33%).

Thermal Demagnetization—Diabase

Extensive thermal demagnetization experiments were carried out on 53 specimens (44 Kaminak diabase, and 9 lamprophyre). These consisted of measuring the specimens after heating them in 50 °C steps from 125 °C to 575 °C: in some cases a 550 °C step was also included. The heating and cooling was carried out in a field of less than 5 gammas and a nitrogen atmosphere was used in the work involving 30 of the specimens. The choice of samples to be thermally demagnetized was based first on the results of AF demagnetization. The initial thermal results formed the basis for additional heating experiments.

Fifteen specimens from the three unmetamorphosed sites (63, 83, 84) were examined and Fig. 4A gives representative normalized demagnetization curves. The magnetization of 12 of these duplicated the AF results by very gradually and continuously changing from a steeply south to a shallow southwesterly direction. This directional change continued steadily up to 475 °C in all cases, and in some cases to 575 °C. Three of the specimens exhibited erratic directional changes during the step heating. However, all of these specimens were heated in air, so it is thought that this behavior was due to oxidation during heating.

Detailed thermal demagnetization was carried out on site 75 (Fig. 4B), because the AF demagnetization curve for this site closely resembled the AF curves for the unaltered sites (Fig. 5B). The rock from site 75 is megascopically rather fresh and after AF cleaning at 200 Oe, the site directions were streaked between the altered and unaltered diabase direction (Fig. 6A) with most samples exhibiting a high θ. For this reason, results from this site could not be used in the AF group statistics. The samples from this site responded better to thermal treatment (Fig. 6B) and in all but sample 6, approached the unaltered diabase direction shown by sites 63, 83, and 84. It is clear that thermal cleaning in this case was more effective than was AF cleaning in removing the effects of secondary components resulting from low-grade metamorphism. At 525 °C the directions of the cores were close to that obtained from the unaltered dikes after optimum AF cleaning, but after the next step at 575 °C, the directional changes became random. Note that intensities dropped sharply between 525 °C and 575 °C (Fig. 4B).

The intensity of magnetization of cores from sites 85 and 86 dropped sharply above 225 °C (Fig. 4C). Both sites contained cores which were stably magnetized in the altered diabase direction after optimum AF cleaning, but both sites included cores which had anomalous directions, some of which changed towards the unaltered end point in response to AF cleaning. Thermal cleaning did not change those exhibiting the

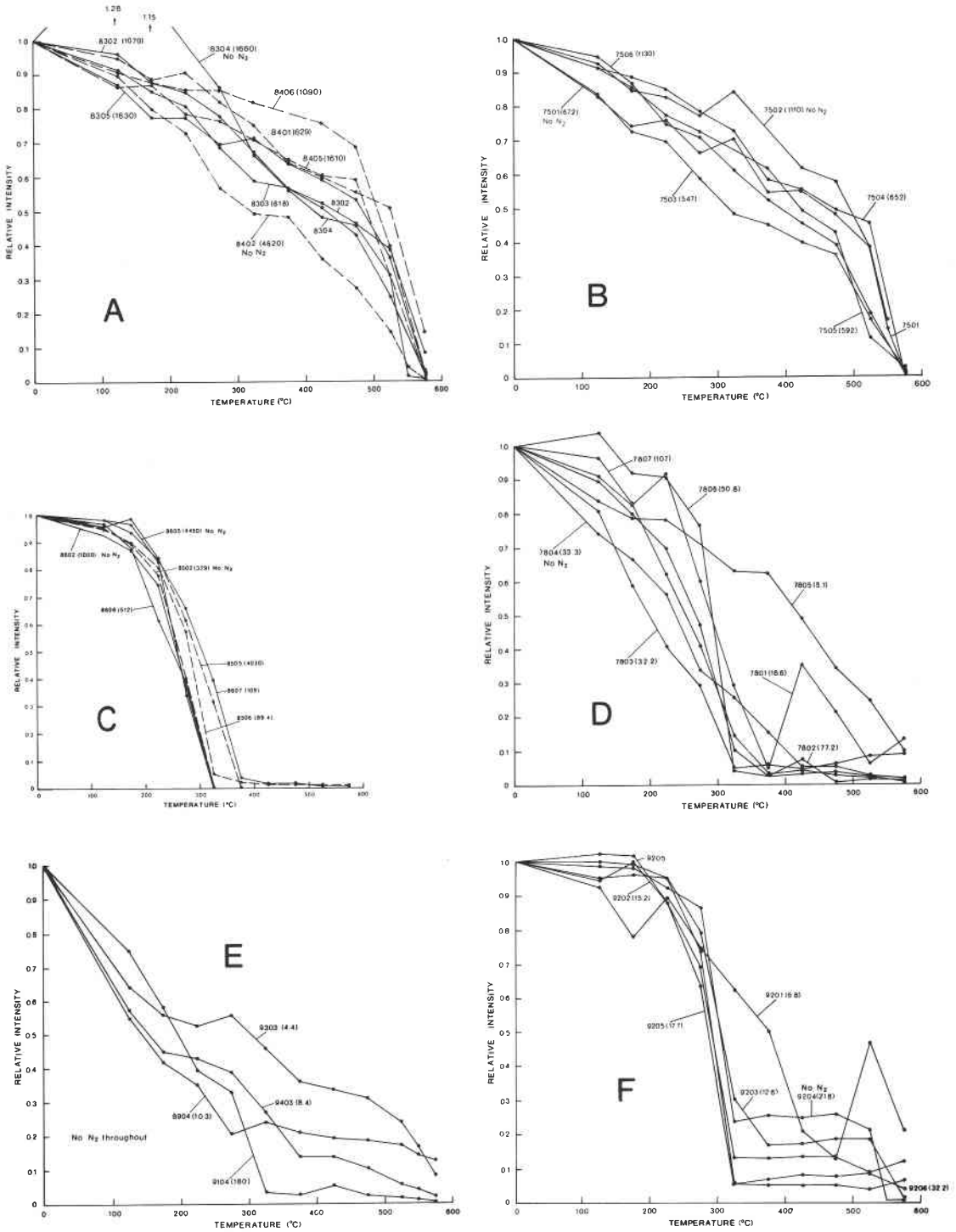


FIG. 4. Normalized thermal demagnetization curves for Kaminak diabase and lamprophyres. Initial intensity of magnetization (in units of 10^{-6} emu/cm^3) are shown in brackets after sample numbers. A: Sites 83 and 84 (unaltered diabase). B: Site 75. C: Sites 85 and 86 (altered diabase). D: Site 78 (altered diabase). E: Samples from 4 lamprophyres. F: Site 92 (lamprophyre).

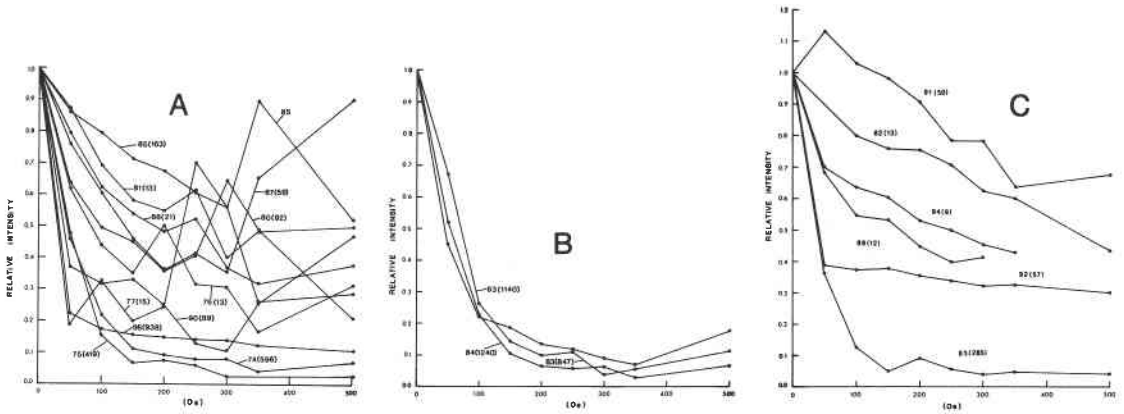


FIG. 5. A: Normalized AF demagnetization curves from Kaminak diabase, altered. B: Normalized AF demagnification curves from Kaminak diabase, unaltered. C: Normalized AF demagnetization curves from Kaminak lamprophyre dikes.

unaltered directions, nor did it produce any end points with those samples that had anomalous AF directions.

Five out of seven cores of site 78 proved to be unstable during AF cleaning, but the directions of specimens from all 7 cores of this site were well grouped after thermal cleaning at 275 °C (Fig. 4D). Their mean direction moved about 10° farther away from the recent earth's field direction with this treatment, and the mean agreed well with the direction obtained for the altered sites under optimum AF cleaning.

In general the altered diabases began to show unstable behavior (sharp intensity drop, rapid irregular directional changes) between 225 °C and 325 °C (Fig. 4C,D). The unaltered dikes were generally stable up to 475 °C and some were stable to 575 °C.

Thermal Demagnetization—Lamprophyre

Thermal demagnetization curves for 5 of the 6 lamprophyre sites are given in Fig. 4E,F. Demagnetization of test specimens from 4 of the 5 sites are shown in Fig. 4E. These curves indicate an exponential decrease in intensity with heating, and suggest the presence of a very low-temperature component. The curves for the specimens at site 92 (note that 9201 is from a sample of diabase near the lamprophyre) indicate the presence of two distinct components. The changes in direction of these specimens during thermal cleaning is highly instructive (Fig. 7). All of the cores retained a metamorphic diabase dike direction to 275 °C. At 325 °C they shifted suddenly. The direction of core 6 at 325 °C was almost reversed relative to its direction at 275 °C and the remaining cores at 325 °C shifted towards this direction to varying degrees. At higher temperatures they continued to move towards the reversed direction (Fig. 7). This is interpreted as indicating the presence of an earlier high temperature reversed component. If this component was formed at the time of lamprophyre intrusion, the intrusion must have taken place at a time of field reversal long after the intrusion of the diabase dikes. Note that AF cleaning of site 92 up to 800 Oe did not reveal the presence of this high temperature earlier component; specimens exhibited only a metamorphic diabase direction.

The directions of test specimens from the other four lamprophyre sites (89, 91, 93, 94) indicate stable end points in the temperature range 275 °C to 575 °C. In this range they have the same direction as the metamorphosed diabase.

Summary and Discussion of Results

Kaminak paleomagnetic data—local significance

The mean pole positions and related statistics for the altered and unaltered diabase dikes and for the lamprophyre dikes are given in Table 2. In all three cases, the mean magnetization directions shifted away from the recent earth's field direction as a result of AF cleaning. There is an exact correlation between the alteration as petrographically determined, and the K–Ar whole-rock ages and paleomagnetic pole positions of the diabase. The altered diabases (22

TABLE 2. Comparison of mean pole positions[1] of diabase and lamprophyre dikes before and after rejecting data from 'unstable' samples

	No. of sites	No. of samples	NRM				Cleaned at 200 Oe			
			Long	Lat	k	α	Long	Lat	k	α
			Kaminak Diabase Dikes (metamorphosed)							
All samples	25	173	90.5° W	39.4° N	8.3	9.8	96.7° W	22.1° N	20.2	6.2
Divergent samples[2] excluded (6)	25	167	88.2° W	40.3° N	10.1	8.8	96.2° W	22.3° N	22.6	5.9
Divergent and inhomogeneous[3] samples excluded (79)	22	94	85.6° W	37.9° N	7.1	11.3	92.1° W	20.4° N	24.3	6.1
			Kaminak Diabase Dikes (unmetamorphosed)							
All samples	3	21	111.1° W	21.5° N	21.0	17.7	124.2° W	21.5° S	36.4	13.4
Divergent samples excluded (1)	3	20	112.4° W	21.6° N	21.3	17.6	125.6° W	22.8° S	29.0	15.1
Divergent and inhomogeneous samples excluded (8)	3	13	117.0° W	22.0° N	33.1	14.1	122.3° W	23.8° S	24.0	16.5
			Kaminak Lamprophyre Dikes							
All samples	6	38	107.0° W	58.6° N	7.3	21.1	94.4° W	39.6° N	17.3	13.8
Divergent and unhomogeneous samples excluded (7)	6	31	106.0° W	58.5° N	6.7	22.1	92.8° W	40.9° N	11.3	17.1

[1]Mean pole positions were all calculated from site poles.
[2]Divergent samples are those whose directions were 3 standard deviations or more from the site mean direction after cleaning at 200 Oe.
[3]Inhomogeneously magnetized samples are those exhibiting $\theta > 25°$ after cleaning at 200 Oe. The data from 3 sites which contained only 1 acceptable sample according to divergence and inhomogeneity criteria were also excluded.

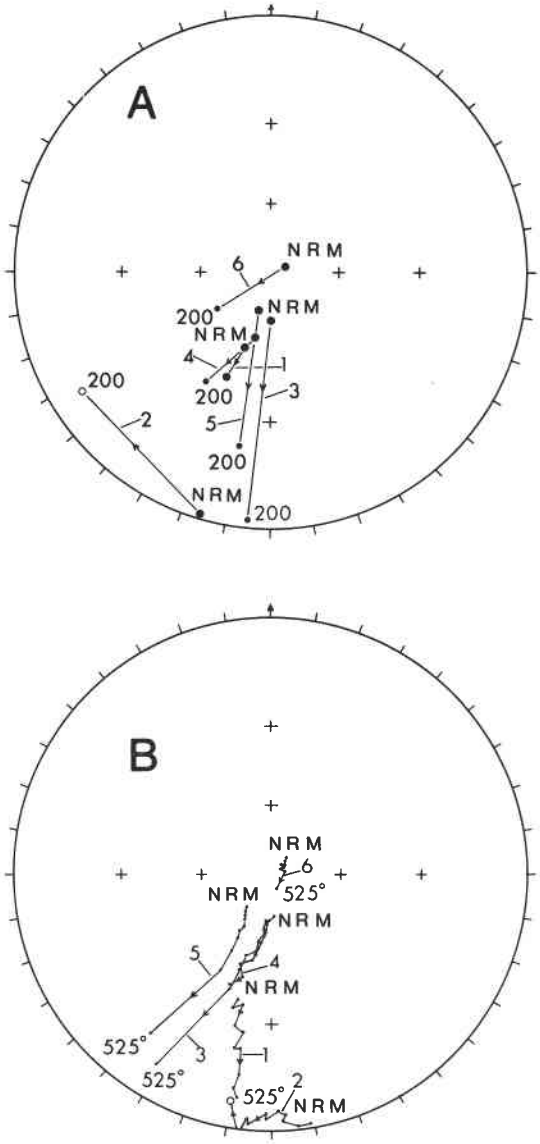


FIG. 6. Magnetization directions for site 75. A: AF demagnetization—average of 2 specimens for each sample. Large dots—$\theta < 25°$; small dot—$\theta > 25°$; open circle—directed upward. B: Thermal demagnetization, one specimen for each sample.

sites) give a pole at 092° W, 20° N, $\alpha_{95} = 6°$ and the three unaltered dikes, a pole at 122° W, 24° S, $\alpha_{95} = 16°$. The latter is based on fewer data, so is not as well defined. It is clear, however, that the pole for the unmetamorphosed dikes must be the older of the two. The initial (NRM) intensity of magnetization of the unaltered dikes is about four times that of the altered. This difference is the result of low-grade regional metamorphism. The intensity of the unaltered dikes drops sharply during AF step demagnetization (Fig. 5B) and at the optimum cleaning field of 200 Oe, the intensities of the unaltered dikes is about twice that of the altered (Table 1). A few of the specimens of altered dikes acquired components as a result of cleaning in higher fields (Fig. 5A). The cause of the effect is not known. The NRM intensity of the lamprophyre dikes is much lower than that of either the unaltered or altered diabase (Table 1). Most of the lamprophyres show a gradual decrease in intensity in response to AF cleaning (Fig. 5C). The magnetic intensity of the lamprophyres is also considerably less than that of the diabase at the optimum field of 200 Oe. The mean poles of the metamorphosed dikes and lamprophyres are separated by about 21 degrees of arc (Fig. 8). However, their circles of confidence overlap.

There are notable differences in trend of the diabase dikes in the vicinity of Kaminak Lake, those to the northwest of the lake having a trend of about 030° and those to the southeast of Kaminak a trend either close to 0° or in some cases to the northwest. If the difference in trend is due to post-consolidation relative movement, this occurred prior to the remagnetization of the dikes,

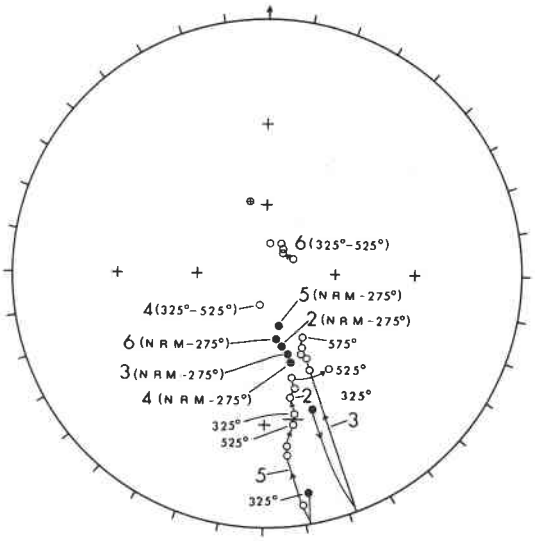


FIG. 7. Magnetization directions of samples from site 92 on thermal demagnetization to 525 °C. Dots—down; open circles—up; circled cross—mean of the 5 samples (NRM—275 °C) reversed.

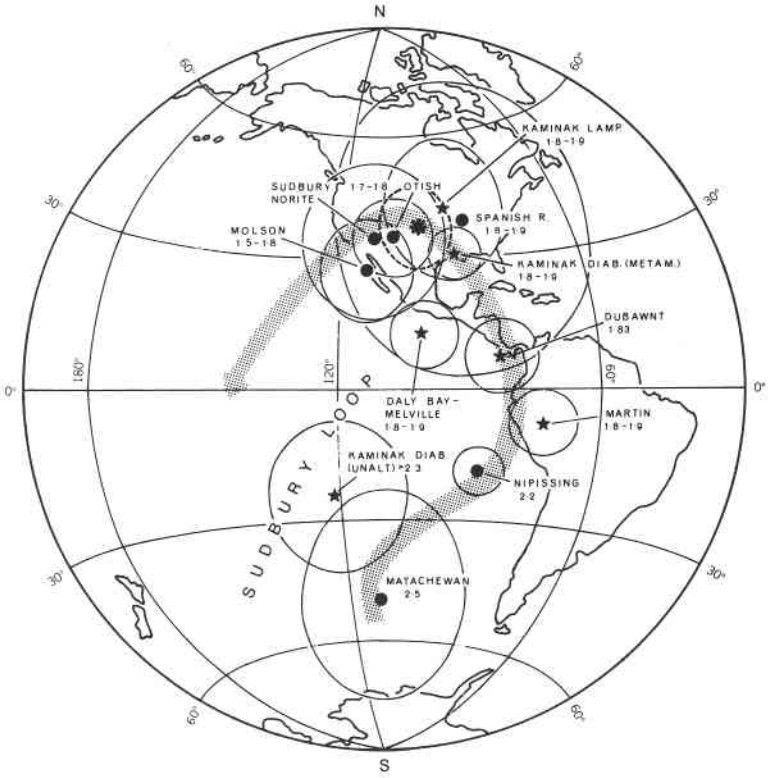


FIG. 8. The polar wandering curve for the Superior Orogen in the period from about 2.5 to 1.7 Ga ago. The number attached to each pole is the probable age of magnetization of the unit, and in some cases, is significantly less than the primary age of the unit. Poles from the Superior Orogen are shown as dots, and from the Churchill and Kaminak Orogens, as stars. The large asterisk (with circle of confidence dashed) is the mean of the six poles near the apex of the Sudbury Loop.

as the metamorphosed dikes show no paleomagnetic evidence for such rotation.

Kaminak paleomagnetic data—regional significance

The purpose of this section will be to discuss the significance of the Kaminak poles in terms of Precambrian polar wandering. In doing so, we thought it best at the beginning to restrict discussion to rock units in this general part of the Churchill Orogen, namely the area from Lake Athabasca to Wager Bay (Fig. 9). This avoids dealing with the assumption that there has been no relative movement between this part of the Churchill and more distant blocks since the Kaminak rocks were magnetized. There are paleomagnetic data available from 3 other areas within the block under consideration, and the ages of magnetization of the rocks of these three areas are probably comparable with the age of the secondary (metamorphic) magnetization of the Kaminak dikes. The three areas are Martin Lake (Evans and Bingham 1973), Dubawnt (Park *et al.* 1973), and Daly Bay – Melville Peninsula (Park 1973). The data for these areas are summarized in Table 3, Part A and on Fig. 8, along with data from Kaminak. We are dealing here with two sorts of data—two sets (the Martin and Dubawnt) are based on volcanic and sedimentary strata, and these data presently reflect magnetization that is primary and acquired at the time of deposition or soon after. The Daly Bay and Kaminak dike results (except for the 3 unmetamorphosed dikes) were obtained from rocks that were remagnetized long after these rocks were formed. In the case of the Kaminak area, all of, and in the case of the Daly Bay, most of, the samples were derived from metamorphosed basic igneous rocks. The metamorphosed Kaminak dikes have no dis-

TABLE 3. Pole positions

Formation	Pole position		α_{95}	Probable age of magnetization (Ma)
A. Poles from Churchill Orogen				
[1]Martin	073° W	08° S	9°	1800–1900
Daly Bay	101° W	12° N	7°	1800–1900
Dubawnt	083° W	07° N	8°	1828 ± 4
Kaminak Diabase (metamorphosed)	092° W	20°N	6°	1800–1900
Kaminak Diabase (unaltered)	122° W	24° S	16.5°	>2300
Kaminak Lamprophyre	093° W	41° N	17°	1700–1800
B. Poles from Superior Orogen				
[2]Spanish River	088° W	38° N	35°	1600–1900
[3]Nipissing Diabase	088° W	19° S	5.4°	2200
[4]Matachewan Diabase	110° W	49° S	27°	2500
Otish Gabbro	107° W	35° N	9°	
Sudbury (see text)	112° W	34° N	19°	1700–1800
Molson	114° W	28° N	12°	1500–1800

[1]Pole position based on site mean magnetization directions given by Evans and Bingham (1973) as 072° W, 09° S, dp = 6.3°, dm = 11.5°.
[2]Pole position based on sample magnetization directions given by Robertson and Watkinson (1974) as 096° W, 37° N, dp = 24°, dm = 25°.
[3]Pole position based on site mean magnetization directions given by Symons (1970) as 92° E, 19° N, dp = 3.9°, dm = 6.5°.
[4]Mean of 3 pole positions—one normal and one reversed from Fahrig *et al.* (1965), and one from 4 sites of Strangway (1961).

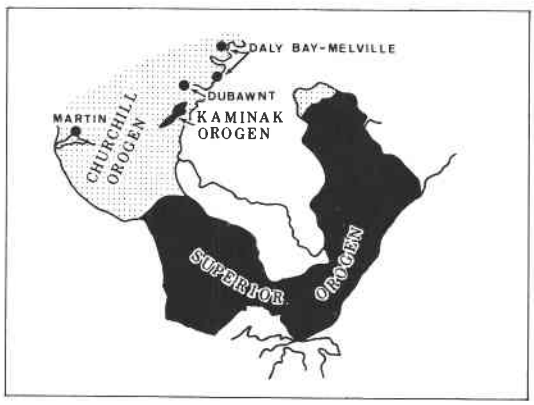


FIG. 9. The relationship between the Superior, Kaminak, and Churchill Orogens and the locations of areas within the eastern Churchill Orogen for which reliable paleomagnetic data are available.

cernible metamorphic rock fabric, whereas the Daly Bay rocks are strongly foliated (Park 1973).

The Dubawnt volcanics have yielded a whole-rock Rb/Sr age of formation (and presumably of magnetization) of 1725 (1828) ± 4 Ma (Wanless and Loveridge 1972). There is no direct evidence, however, for the age of volcanism and sedimentation of the Martin Formation. Fraser *et al.* (1970) suggested that a whole-rock K–Ar age of 1830 Ma from a basic dike overlain unconformably by Martin Formation provided a maximum age for the Martin. However, K–Ar whole-rock ages tend to be low because of argon loss. The Rb/Sr age of 1860 (1975) ± 20 Ma for discordant pegmatites (Sassano *et al.* 1972) in the Martin Lake area, may provide evidence of the late stages of Hudsonian regional metamorphism and provide an upper limit for deposition of the Martin. Sassano *et al.* (1974) have subsequently obtained a Rb/Sr isochron age of 1675 (1780) ± 72 Ma on lamprophyre dikes that cut the pre-Martin basement. These dikes are probably pre-Martin (L. P. Tremblay, personal communication 1974), and provide a maximum age for Martin volcanism and sedimentation. It seems likely that the Martin was deposited about 1800 Ma ago.

It is more difficult to establish a probable age for the magnetization of the metamorphosed Kaminak dikes and for the metamorphic rocks of the Daly Bay area. Park (1973) suggested an age of magnetization for the Daly Bay rocks of 1640 Ma, in other words, essentially the same age as that given by K–Ar determinations on biotites extracted from the igneous and metamorphic rocks of this region (12 determinations, range 1580 to 1690, mean 1622; Heywood *in* Wanless *et al.* 1967). This seems unlikely. To the

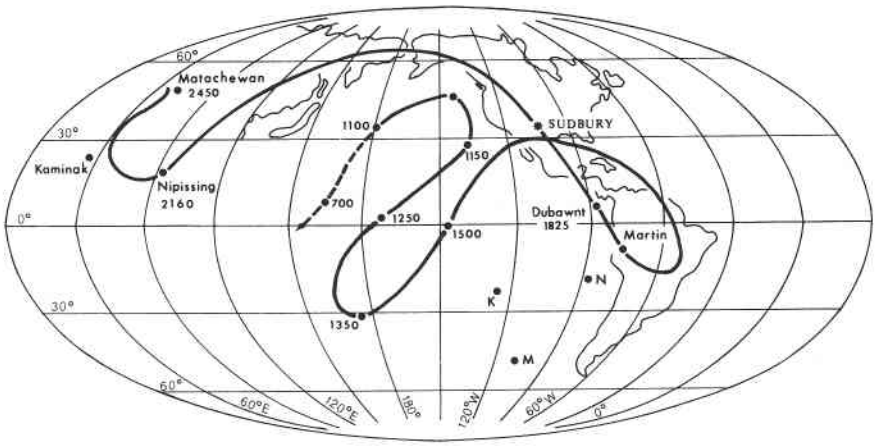


FIG. 10. Polar wandering curve for the Superior Orogen, assuming that the Matachewan, Kamanak diabase (unmetamorphosed), and Nipissing have antipodal positions relative to those shown on Fig. 8.

west the Dubawnt volcanics lie with profound unconformity on gneisses of the Churchill Orogen (Fraser *et al.* 1970), so unless the Daly Bay area was uplifted and cooled much later than was the Dubawnt area, the age of the Dubawnt volcanics (1725 (1828) ± 4 Ma) provides the minimum age for the magnetization of the Daly Bay rocks. It is interesting to note that if this is the case the lowest K–Ar biotite age of that region (1580 Ma) is at least 250 Ma less than the age of remanent magnetization. It is most probable then that the magnetization reported by Park is between 1800 and 1900 Ma old.

The age of magnetization of the metamorphosed Kaminak dikes poses a similar problem. The time of intrusion is known from K–Ar ages to be early Aphebian, probably more than 2300 Ma ago, and the lowest K–Ar ages from within the low-grade metamorphic belt are 1690 Ma (Davidson 1970). It is quite likely that the age of magnetization of the metamorphic Kaminak dikes is at least 1800 Ma. The three unmetamorphosed dikes (sites 63, 83, 84) are thought to have been magnetized at the time of their intrusion more than 2300 Ma ago.

The magnetization of the lamprophyre dikes is somewhat puzzling. They have been described as post-tectonic by Davidson (1970) inasmuch as they are undeformed and cut folded Hurwitz (Aphebian) strata. On the basis of a K–Ar age of 1690 ± 55 Ma, obtained on biotite from the lamprophyre at site 94, he suggested that they might be related to Dubawnt igneous activity. They are known to cut metamorphosed Kaminak diabases (for example, site 82). A K–Ar determination on a mica from lamprophyre site 82 has yielded an age of 2071 ± 47 Ma. If this is the minimum age of intrusion it is significantly greater than the age of Hurwitz volcanism (1808 (1916) ± 35) reported by Wanless and Eade (1975) or of Dubawnt volcanism which is (1725 (1828) ± 4). We propose in explanation that the 2071 Ma K–Ar age for the biotite of the lamprophyre of site 82 is anomalously high, because of the presence of excess argon. This area is underlain by Archean gneisses from which argon in large quantities would be released during the Hudsonian Orogeny. It is quite possible that the biotite-rich lamprophyre absorbed significant quantities of this fugitive argon from the country rocks (Stockwell 1963; Wanless *et al.* 1969). It is suggested that the lamprophyres have a pole position similar to that of the metamorphosed Kaminak because they were intruded at the time of the Sudbury cross-over, at least 100 Ma after the metamorphism and magnetization of the Kaminak dikes (Fig. 10).

Kaminak paleomagnetic data—plate tectonic significance

It is the purpose of this section to discuss the Kaminak data in relation to results from the Superior orogen. This is in line with the suggestion that the Superior orogen, including rocks that lie undisturbed on the Superior orogen, be

used as reference plate for discussing Precambrian North American paleomagnetic results (Roy and Fahrig 1973). A crucial question in considering data from the Superior orogen is whether a north-trending loop in the polar wandering path, the 'Sudbury loop' (Fahrig and Chown 1973) is indicated by rocks of the orogen magnetized during the period approximately 2500 Ma to 1500 Ma ago. The apex of this loop is defined by the pole for the north range of the Sudbury norite (Sopher 1963; Larochelle 1969). The Sudbury norite pole used in Table 3 and Fig. 8 was obtained by combining Sopher and Larochelle's data in such a way that norite data from each sector in the north range was considered to be a single site and data from micropegmatite of these sectors, separate sites. However, the north range has probably been tilted (originally demonstrated paleomagnetically by Hood 1961) and, following Sopher, a corrected pole position was calculated by assuming a post-magnetization tilt of 10°. The pole position thus obtained is unlikely to be in error by more than 10° of arc. Other results from the Superior orogen have tended to support the existence of the northern extension of the loop. These data are given in Table 3, part B, and the pole positions are shown on Fig. 8. The Molson pole is data (Ermanovics and Fahrig 1975) from diabase dikes that lie well within the Superior. These dikes exhibit a reversal and their radiometric age may indicate that their pole is related to the descending west limb of the Sudbury loop.

The progression of poles from rocks of the Churchill orogen (shown as star symbols on Fig. 8) from the unaltered Kaminak diabase dikes to the Kaminak lamprophyre is in remarkably good agreement with the general path suggested by Superior rocks. Their K–Ar age, trend, and even their porphyritic texture, suggests that the Kaminak dikes may be correlatives of the Matachewan dikes, and their unmetamorphosed pole is close to that of the Matachewan. In each orogen, these dikes cut Archean supracrustal rocks whose main structural grain is east–west. It is therefore inferred that there has been no large-scale shift or rotation of the Kaminak orogen relative to the Superior since the Matachewan and Kaminak dikes were intruded.

There is, however, a possible stratigraphic problem involved in the curve as here shown. As previously mentioned, the Martin and Dubawnt lie unconformably on gneisses that were involved in thermal events of the Hudsonian orogeny. Presumably the magnetization of the metamorphosed Kaminak and of the Daly Bay rocks dates from these thermal events. The Kaminak and Daly Bay poles should therefore pre-date the Martin and Dubawnt, but according to Fig. 8, this is not the case. There are several possible explanations for this. It may be, as suggested by Piper *et al.* (1973), that the older North American poles, *e.g.*, the Matachewan, Kaminak diabase (unmetamorphosed), and necessarily the Nipissing as usually plotted, are actually antipodal points in terms of the polar wandering curve. If this is the case, some poles near the top of the Sudbury loop would be older than the Dubawnt and Martin. This would require a very large polar shift between Nipissing diabase time, and the time represented by the cluster of poles at the top of the loop (Fig. 10).

The cluster of 6 poles at the top of the Sudbury loop may be of special interest. Like the Mackenzie pole (Fahrig and Jones 1969), it may represent a quasi-static period in Precambrian polar wandering, or as shown in Fig. 10, it probably represents a cross-over in the Sudbury loop.

Data published after completion of the present paper (Symons 1975) indicate that the Gowganda Formation has a pole position at 123° W, 65° N. This is remarkably close to the polar path of Fig. 10, but the stratigraphic age of these sediments is greater than the Nipissing, whereas the path of Fig. 10 suggests a younger magnetization. It is possible in explanation that the sediments of the collection sites were magnetized after the intrusion (and magnetization) of the Nipissing diabase.

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